

Gradient descent

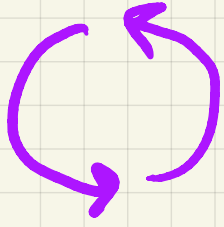
Weekend 1, Friday morning

Carlos Cotrini

Friday 4 September 2026

So far: House price pred. with Robbins-Monro algo
and Lin. reg. model

$$y = w_1 x_1 + w_2 x_2 + \dots + w_d x_d$$

while $|\bar{l}| > \text{tol}$: 

$\hat{y}$ = compute $\leftarrow$ pred from X and w

$$l = y - \hat{y}$$
$$\bar{l} = (l_1 + \dots + l_n) / n$$

$$w = w + \alpha l$$

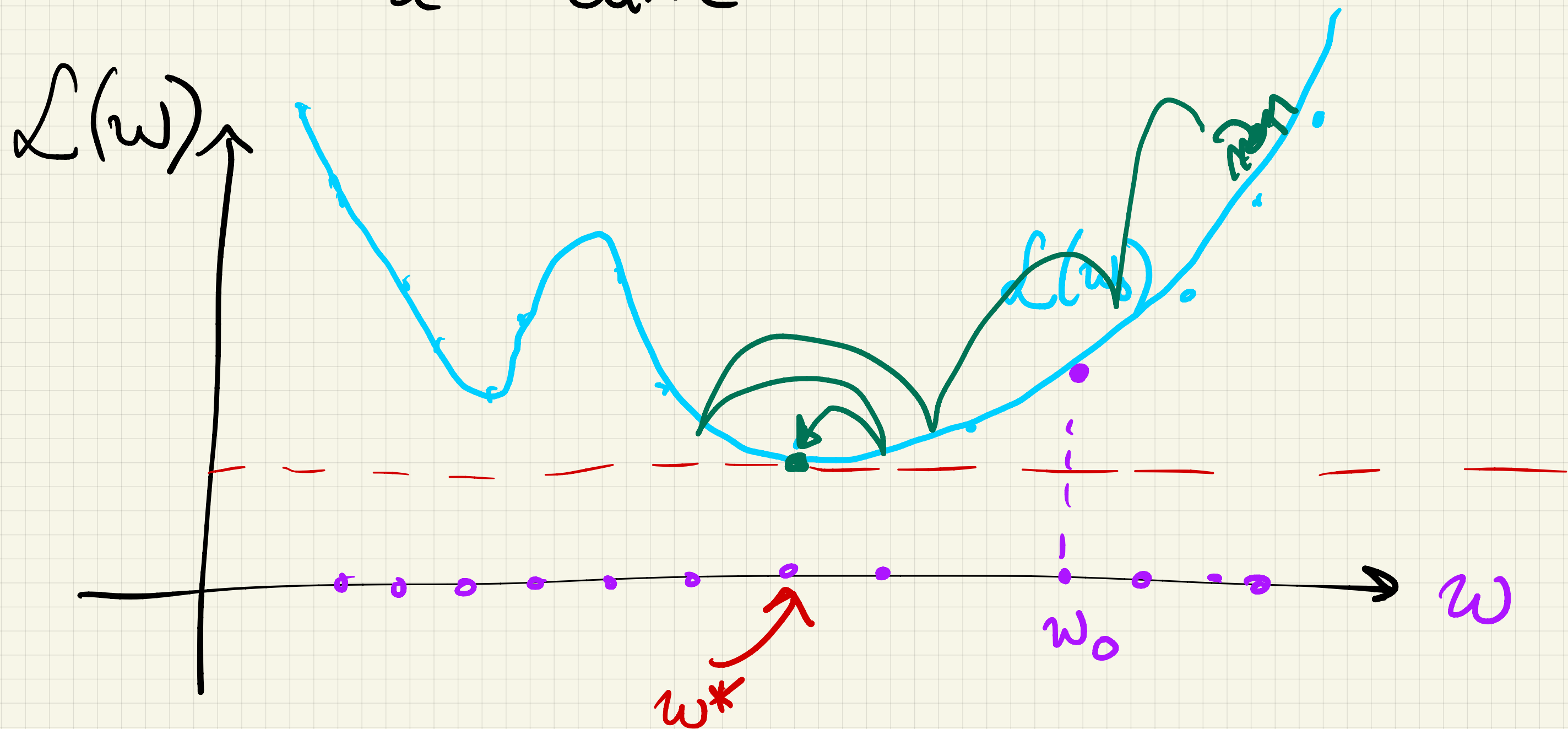
We don't know
we only guess

1

Section 1

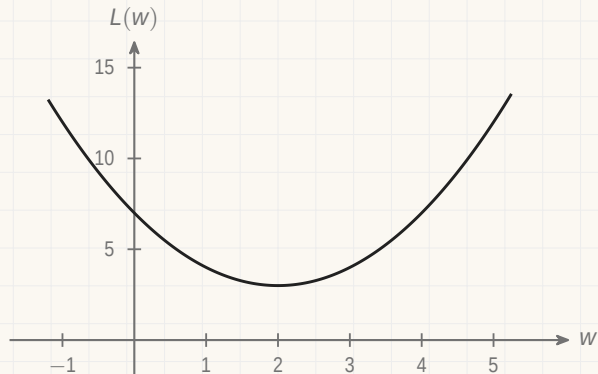
Reading a curve

The loss function can be described w/
a curve



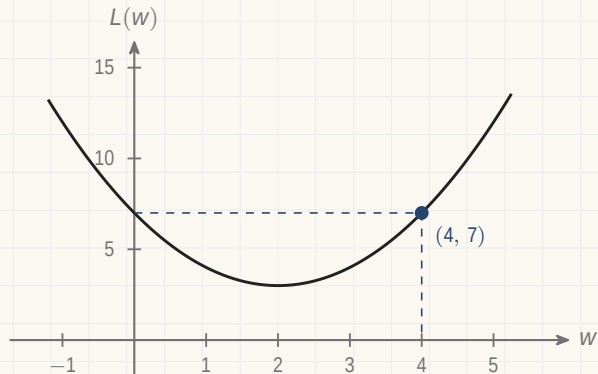
The loss curve

$$L(w) = w^2 - 4w + 7$$



The loss curve

$$L(w) = w^2 - 4w + 7$$



Reading the curve

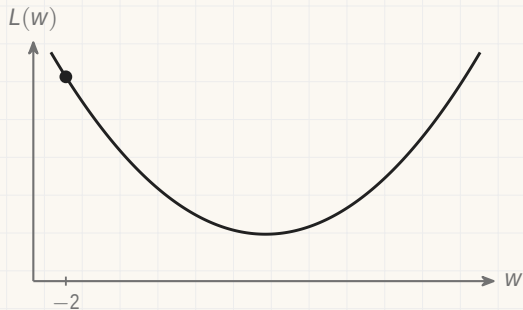
Stand somewhere. Read the height. Decide where to go next.



w	$L(w)$	Update
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Reading the curve

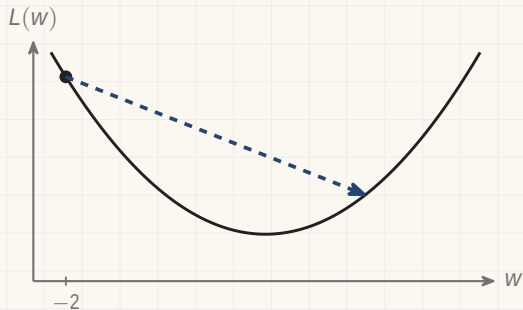
Stand somewhere. Read the height. Decide where to go next.



w	$L(w)$	Update
-2	19	

Reading the curve

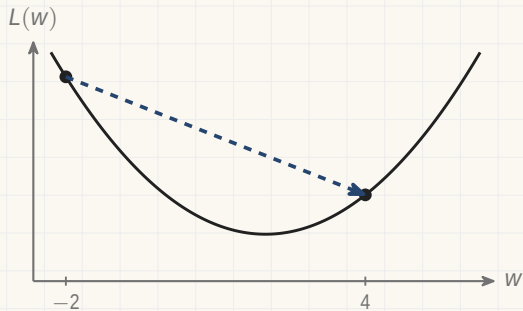
Stand somewhere. Read the height. Decide where to go next.



w	$L(w)$	Update
-2	19	

Reading the curve

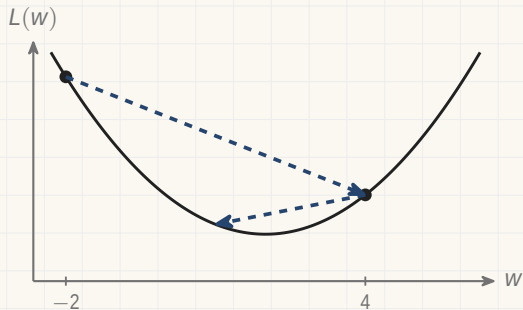
Stand somewhere. Read the height. Decide where to go next.



w	$L(w)$	Update
-2	19	
4	7	

Reading the curve

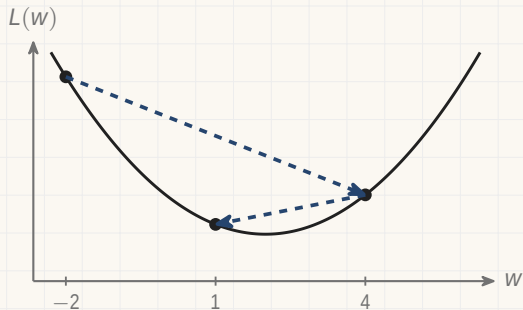
Stand somewhere. Read the height. Decide where to go next.



w	$L(w)$	Update
-2	19	
4	7	

Reading the curve

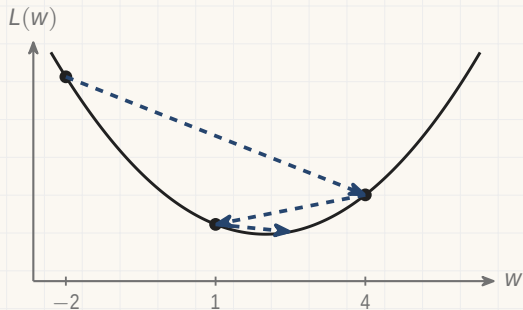
Stand somewhere. Read the height. Decide where to go next.



w	$L(w)$	Update
-2	19	→
4	7	←
1	4	

Reading the curve

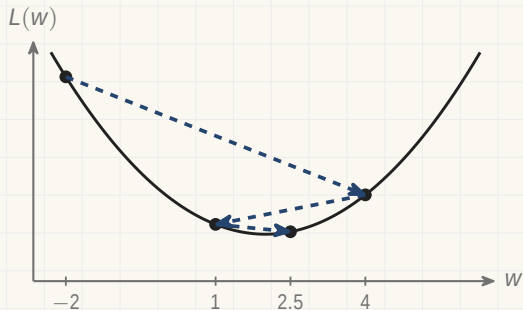
Stand somewhere. Read the height. Decide where to go next.



w	$L(w)$	Update
-2	19	→
4	7	←
1	4	→

Reading the curve

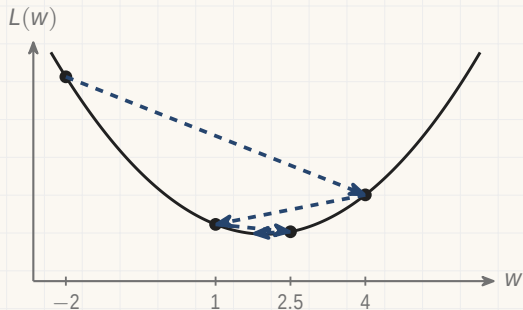
Stand somewhere. Read the height. Decide where to go next.



w	$L(w)$	Update
-2	19	→
4	7	←
1	4	→
2.5	3.25	

Reading the curve

Stand somewhere. Read the height. Decide where to go next.



w	$L(w)$	Update
-2	19	→
4	7	←
1	4	→
2.5	3.25	←

Each arrow is half the length of the one above it, and points the other way.

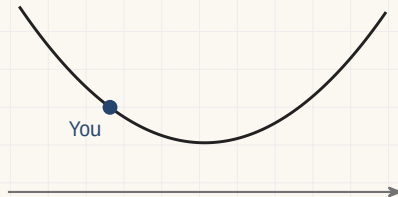
2

Section 2

The grasshopper game

The grasshopper game

You are a grasshopper on a hillside in the dark. You want the lowest point.



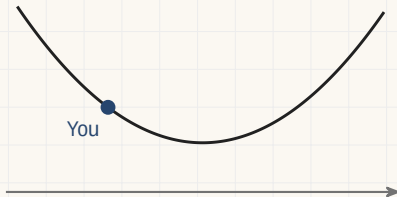
Play it here:

[Grasshopper game address, from the teaching assistants]

The grasshopper game

You are a grasshopper on a hillside in the dark. You want the lowest point.

- You know where you are standing.



Play it here:

[Grasshopper game address, from the teaching assistants]

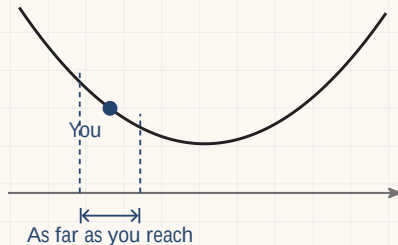
The grasshopper game

You are a grasshopper on a hillside in the dark. You want the lowest point.

- You know where you are standing.
- You can feel the ground under your feet, as far as your legs reach.

Play it here:

(Grasshopper game address, from the teaching assistants)



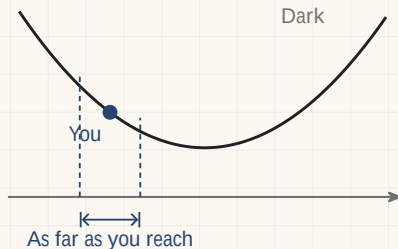
The grasshopper game

You are a grasshopper on a hillside in the dark. You want the lowest point.

- You know where you are standing.
- You can feel the ground under your feet, as far as your legs reach.
- You cannot see the hillside.

Play it here:

(Grasshopper game address, from the teaching assistants)



One round of the game

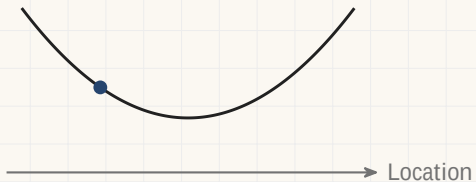
The 09:00 loop again, with the grasshopper's own four words.

Location

Height

Jump

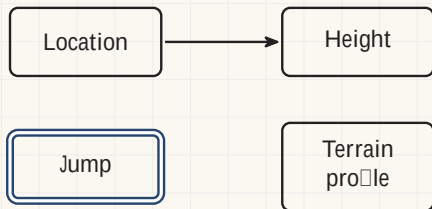
Terrain
profile



One round of the game

The 09:00 loop again, with the grasshopper's own four words.

①

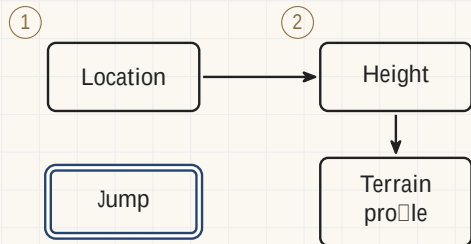


Where you stand, how high it is,
how the ground tilts, where you land

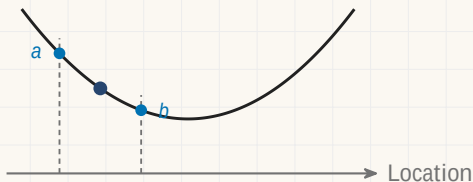


One round of the game

The 09:00 loop again, with the grasshopper's own four words.

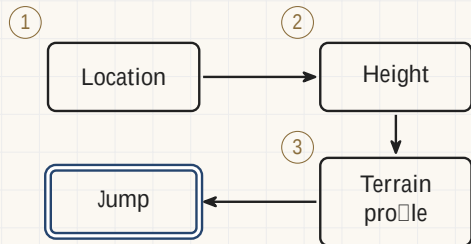


Where you stand, how high it is,
how the ground tilts, where you land

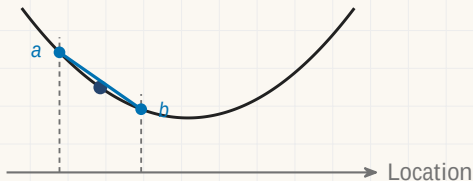


One round of the game

The 09:00 loop again, with the grasshopper's own four words.



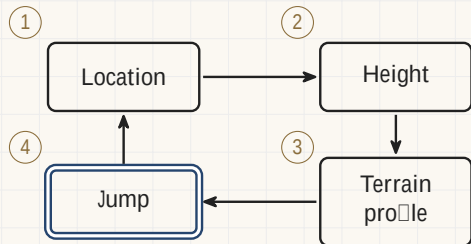
Where you stand, how high it is,
how the ground tilts, where you land



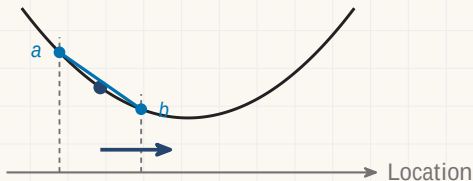
The tilt between the two feet.

One round of the game

The 09:00 loop again, with the grasshopper's own four words.



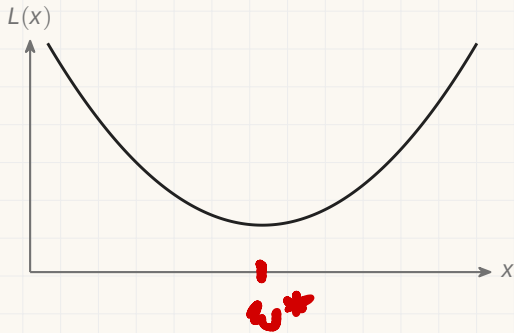
Where you stand, how high it is,
how the ground tilts, where you land



The tilt between the two feet.

The game, written down

$$\begin{array}{ccc} L : & x & \mapsto & L(x) \\ & \uparrow & & \uparrow \\ & \text{Location} & & \text{Height} \end{array}$$



Find w^* s.t.
for any other w

$$L(w^*) \leq L(w).$$

The game, written down

$$\begin{array}{ccc} L : & x & \longmapsto & L(x) \\ & \uparrow & & \uparrow \\ & \text{Location} & & \text{Height} \end{array}$$

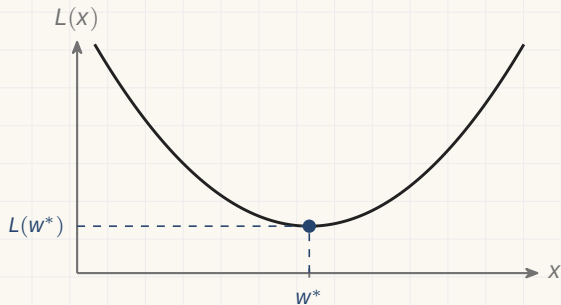
Find w^* in $\mathbb{R}$ such that $L(w^*) \leq L(x)$
for every x in $\mathbb{R}$.



The game, written down

$$\begin{array}{ccc} L : & x & \longmapsto & L(x) \\ & \uparrow & & \uparrow \\ & \text{Location} & & \text{Height} \end{array}$$

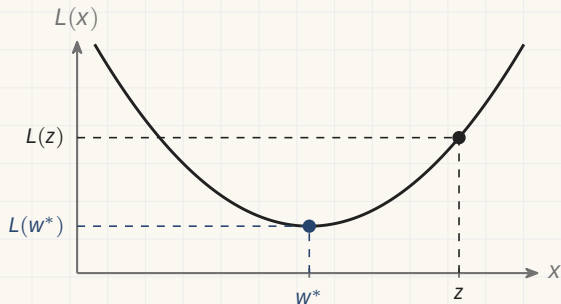
Find w^* in $\mathbb{R}$ such that $L(w^*) \leq L(x)$
for every x in $\mathbb{R}$.



The game, written down

$$\begin{array}{ccc} L : & x & \longmapsto & L(x) \\ & \uparrow & & \uparrow \\ & \text{Location} & & \text{Height} \end{array}$$

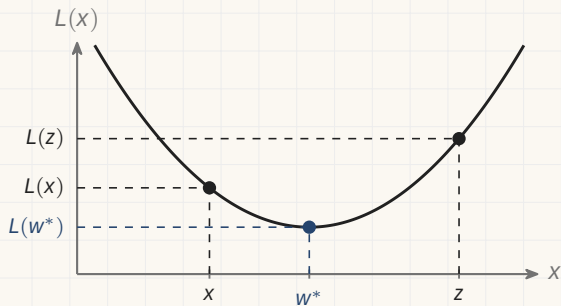
Find w^* in $\mathbb{R}$ such that $L(w^*) \leq L(x)$
for every x in $\mathbb{R}$.



The game, written down

$$\begin{array}{ccc} L : & x & \longmapsto & L(x) \\ & \uparrow & & \uparrow \\ & \text{Location} & & \text{Height} \end{array}$$

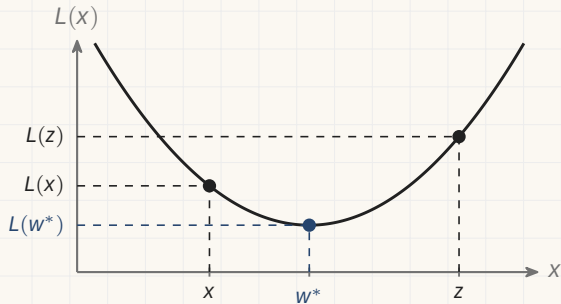
Find w^* in $\mathbb{R}$ such that $L(w^*) \leq L(x)$
for every x in $\mathbb{R}$.



The game, written down

$$\begin{array}{ccc} L : & x & \mapsto & L(x) \\ & \uparrow & & \uparrow \\ & \text{Location} & & \text{Height} \end{array}$$

Find w^* in $\mathbb{R}$ such that $L(w^*) \leq L(x)$
for every x in $\mathbb{R}$.



Nothing in that sentence mentions a hillside.

3

Section 3

Reading the ground under your feet

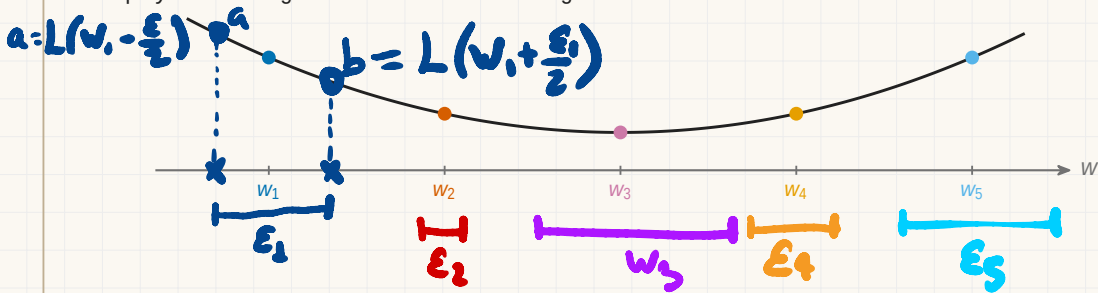
Five players, five windows

Each player feels the ground as far as their own legs reach.



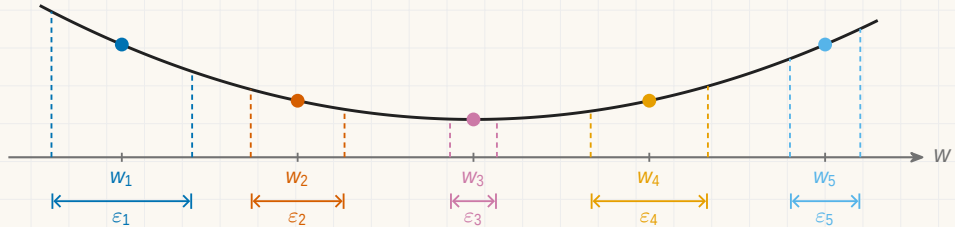
Five players, ∞ ve windows

Each player feels the ground as far as their own legs reach.



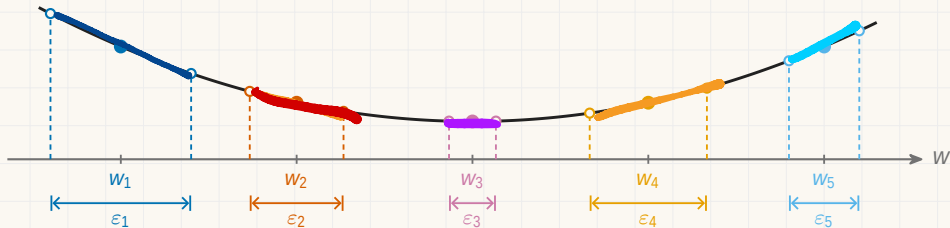
Five players, five windows

Each player feels the ground as far as their own legs reach.



Five players, five windows

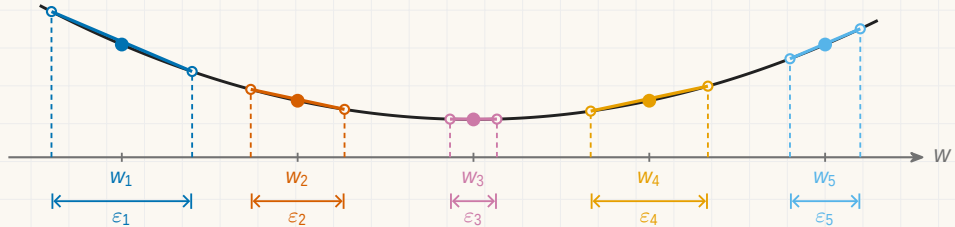
Each player feels the ground as far as their own legs reach.



$$\text{Tilt} = \frac{b-a}{\epsilon}$$

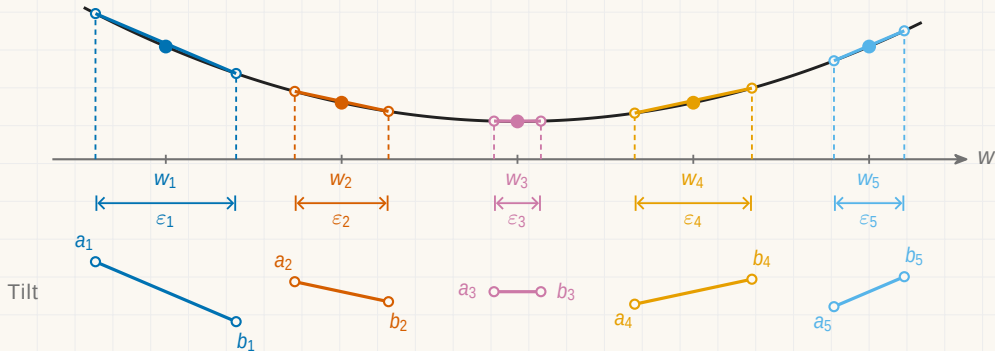
Five players, five windows

Each player feels the ground as far as their own legs reach.



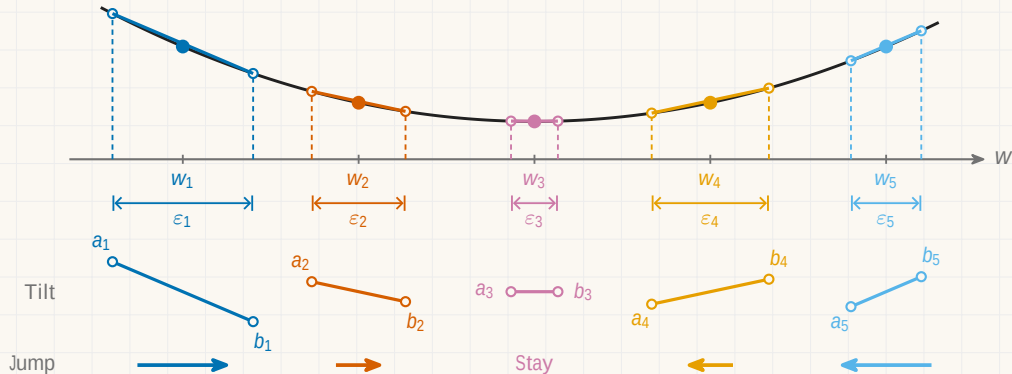
Five players, five windows

Each player feels the ground as far as their own legs reach.



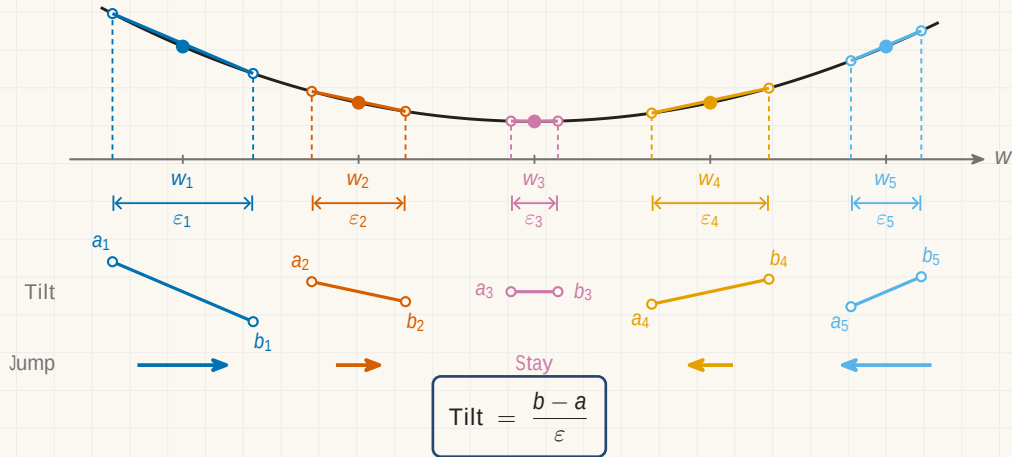
Five players, five windows

Each player feels the ground as far as their own legs reach.

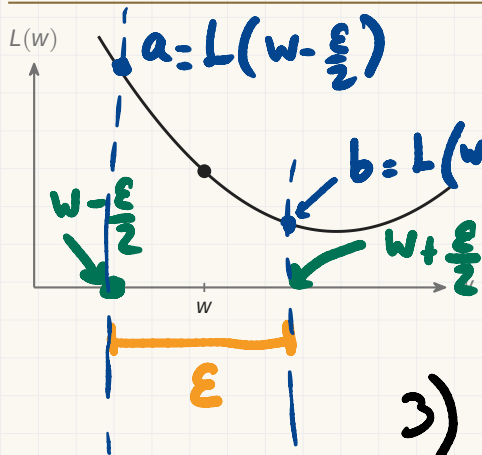


Five players, five windows

Each player feels the ground as far as their own legs reach.



The window, and its name



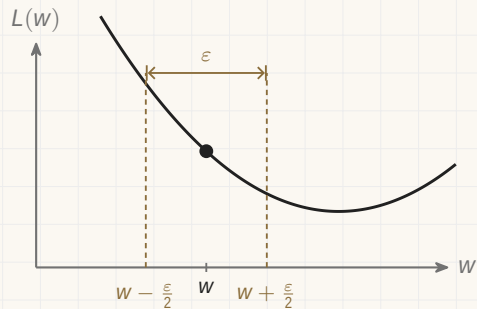
1) Set $\epsilon > 0$

$$a = L(w - \frac{\epsilon}{2})$$

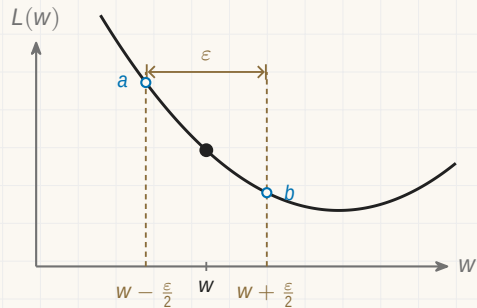
$$b = L(w + \frac{\epsilon}{2})$$

$$3) \nabla_{\epsilon} L(w) = \text{Tilt} = \frac{b - a}{\epsilon}$$

The window, and its name



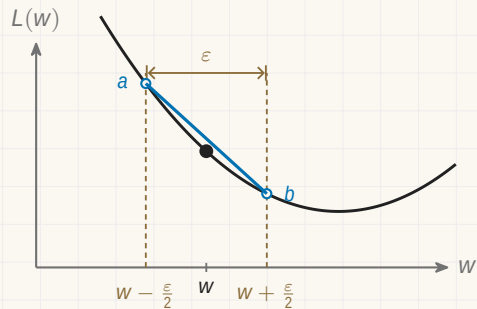
The window, and its name



$$a = L\left(w - \frac{\epsilon}{2}\right)$$

$$b = L\left(w + \frac{\epsilon}{2}\right)$$

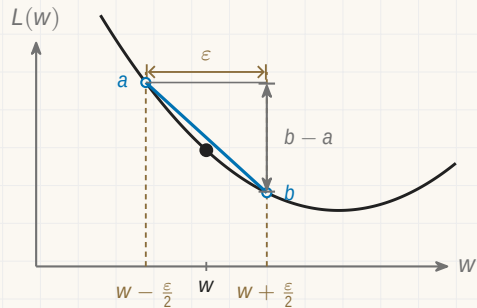
The window, and its name



$$a = L\left(w - \frac{\epsilon}{2}\right)$$

$$b = L\left(w + \frac{\epsilon}{2}\right)$$

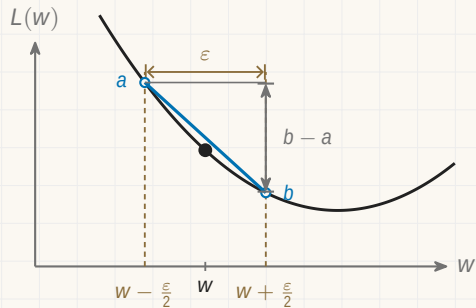
The window, and its name



$$a = L\left(w - \frac{\epsilon}{2}\right)$$

$$b = L\left(w + \frac{\epsilon}{2}\right)$$

The window, and its name

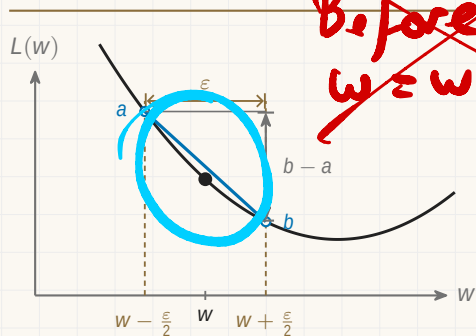


$$a = L\left(w - \frac{\epsilon}{2}\right)$$

$$b = L\left(w + \frac{\epsilon}{2}\right)$$

$$\begin{aligned}\nabla_{\epsilon} L(w) &= \frac{b - a}{\epsilon} \\ &= \frac{L\left(w + \frac{\epsilon}{2}\right) - L\left(w - \frac{\epsilon}{2}\right)}{\epsilon}\end{aligned}$$

The window, and its name



~~Before~~
 ~~$w = w + \alpha h$~~

The new update

Now!

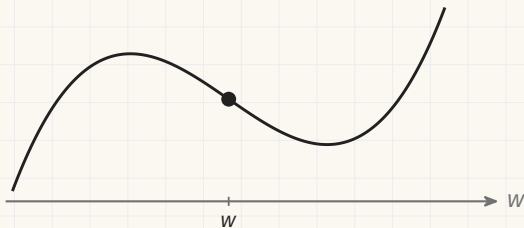
$$w = w + \alpha \nabla_{\epsilon} L(w)$$
$$a = L\left(w - \frac{\epsilon}{2}\right)$$
$$b = L\left(w + \frac{\epsilon}{2}\right)$$

$$\nabla_{\epsilon} L(w) = \frac{b - a}{\epsilon}$$
$$= \frac{L\left(w + \frac{\epsilon}{2}\right) - L\left(w - \frac{\epsilon}{2}\right)}{\epsilon}$$

Its name is the **central difference quotient** of L at w over a window of width ϵ .

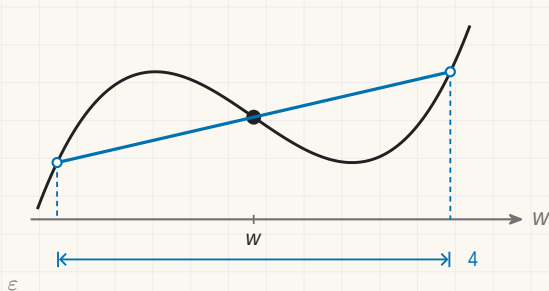
Which window is the right one

A different L , one that is not a bowl, and one place to stand on it.



Which window is the right one

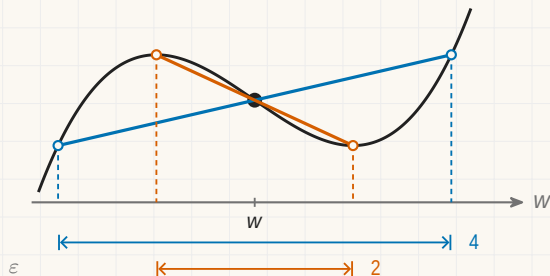
A different L , one that is not a bowl, and one place to stand on it.



$$\varepsilon = 4 \quad \nabla_4 L(w) = 1$$

Which window is the right one

A different L , one that is not a bowl, and one place to stand on it.

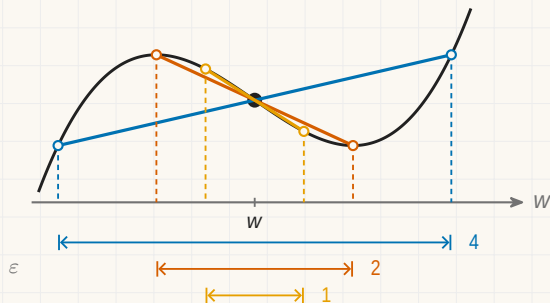


$$\epsilon = 4 \quad \nabla_4 L(w) = 1$$

$$\epsilon = 2 \quad \nabla_2 L(w) = -2$$

Which window is the right one

A different L , one that is not a bowl, and one place to stand on it.



$$\varepsilon = 4 \quad \nabla_4 L(w) = 1$$

$$\varepsilon = 2 \quad \nabla_2 L(w) = -2$$

$$\varepsilon = 1 \quad \nabla_1 L(w) = -2.75$$

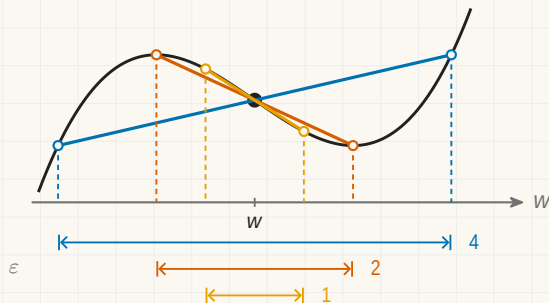
$$\varepsilon = 0.5 \quad \dots \quad -2.9$$

$$\varepsilon = 0.01 \quad -2.999$$

$$\varepsilon = 0.00001 \quad -2.999999$$

Which window is the right one

A different L , one that is not a bowl, and one place to stand on it.



$$\varepsilon = 4 \quad \nabla_4 L(w) = 1$$

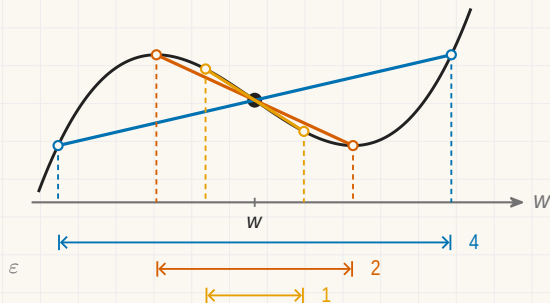
$$\varepsilon = 2 \quad \nabla_2 L(w) = -2$$

$$\varepsilon = 1 \quad \nabla_1 L(w) = -2.75$$

The widest window says the ground rises to the right. It does not.

Which window is the right one

A different L , one that is not a bowl, and one place to stand on it.



$$\varepsilon = 4 \quad \nabla_4 L(w) = 1$$

$$\varepsilon = 2 \quad \nabla_2 L(w) = -2$$

$$\varepsilon = 1 \quad \nabla_1 L(w) = -2.75$$

The widest window says the ground rises to the right. It does not.

$$\varepsilon \rightarrow 0$$



$$\nabla L(w) = -3$$

The gradient

Make the window arbitrarily narrow. For the functions this course meets, the answers settle on one number.

$$\nabla L(w) = \lim_{\varepsilon \rightarrow 0} \frac{L(w + \frac{\varepsilon}{2}) - L(w - \frac{\varepsilon}{2})}{\varepsilon}$$

The gradient

Make the window arbitrarily narrow. For the functions this course meets, the answers settle on one number.

$$\nabla L(w) = \lim_{\varepsilon \rightarrow 0} \frac{L(w + \frac{\varepsilon}{2}) - L(w - \frac{\varepsilon}{2})}{\varepsilon}$$

- It is one number, and it is a tilt: how fast L climbs at w , and in which direction.

The gradient

Make the window arbitrarily narrow. For the functions this course meets, the answers settle on one number.

$$\nabla L(w) = \lim_{\varepsilon \rightarrow 0} \frac{L(w + \frac{\varepsilon}{2}) - L(w - \frac{\varepsilon}{2})}{\varepsilon}$$

- It is one number, and it is a tilt: how fast L climbs at w , and in which direction.
- A negative gradient means the ground falls away to the right, so the lower place is to the right.

The gradient

Make the window arbitrarily narrow. For the functions this course meets, the answers settle on one number.

$$\nabla L(w) = \lim_{\varepsilon \rightarrow 0} \frac{L(w + \frac{\varepsilon}{2}) - L(w - \frac{\varepsilon}{2})}{\varepsilon}$$

- It is one number, and it is a tilt: how fast L climbs at w , and in which direction.
- A negative gradient means the ground falls away to the right, so the lower place is to the right.
- The steeper the tilt, the further the lower place is likely to be.

The gradient

Make the window arbitrarily narrow. For the functions this course meets, the answers settle on one number.

$$\nabla L(w) = \lim_{\varepsilon \rightarrow 0} \frac{L(w + \frac{\varepsilon}{2}) - L(w - \frac{\varepsilon}{2})}{\varepsilon}$$

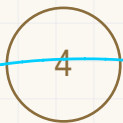
- It is one number, and it is a tilt: how fast L climbs at w , and in which direction.
- A negative gradient means the ground falls away to the right, so the lower place is to the right.
- The steeper the tilt, the further the lower place is likely to be.

Those three lines are the whole of the update rule, and the next frame writes them down.

RM algo
Before
 $w = w + \alpha l$

<

GD
After
 $w = w - \alpha \nabla L(w)$

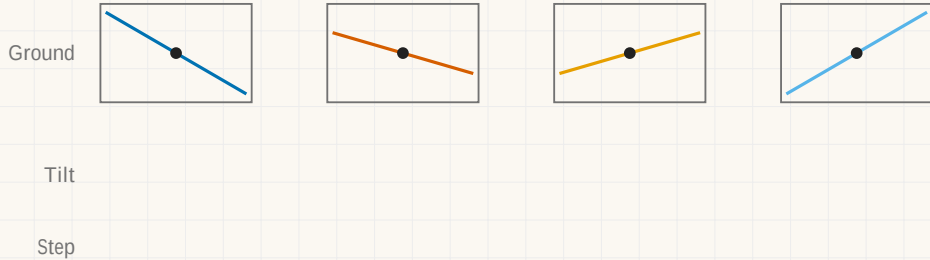


Section 4

Gradient descent

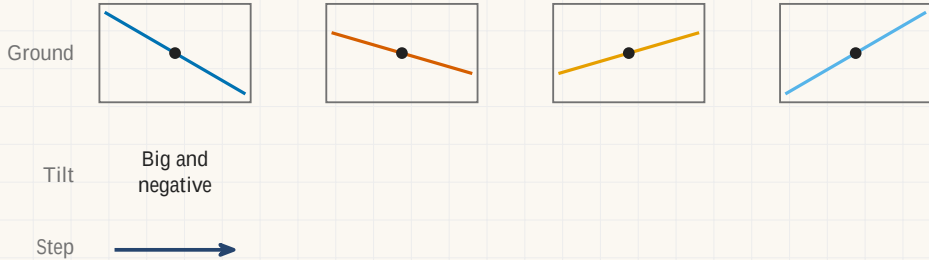
From tilt to step

The same patch of ground, tilted four ways.



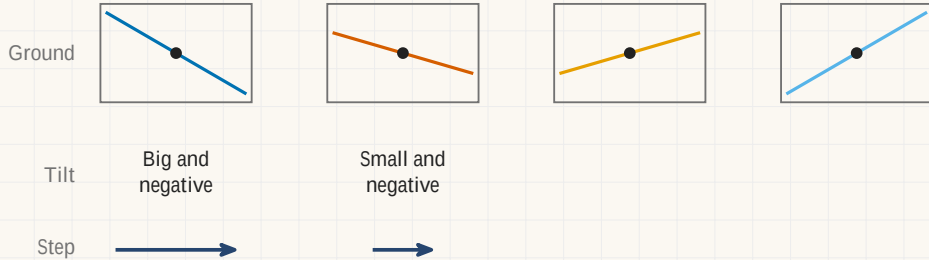
From tilt to step

The same patch of ground, tilted four ways.



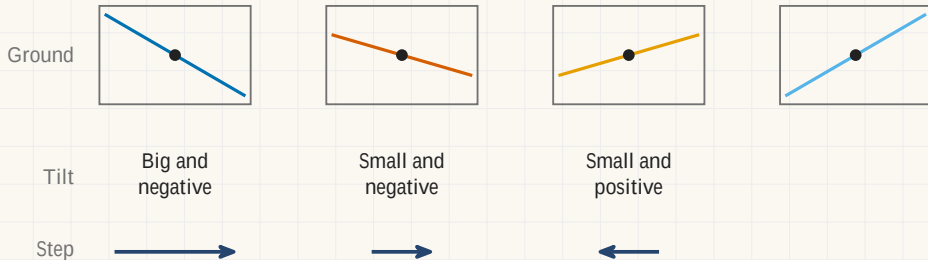
From tilt to step

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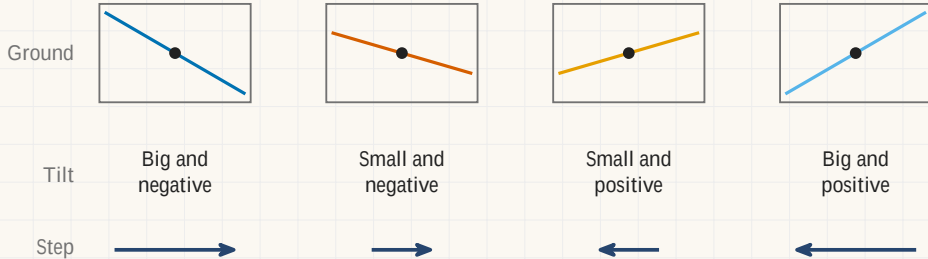
From tilt to step

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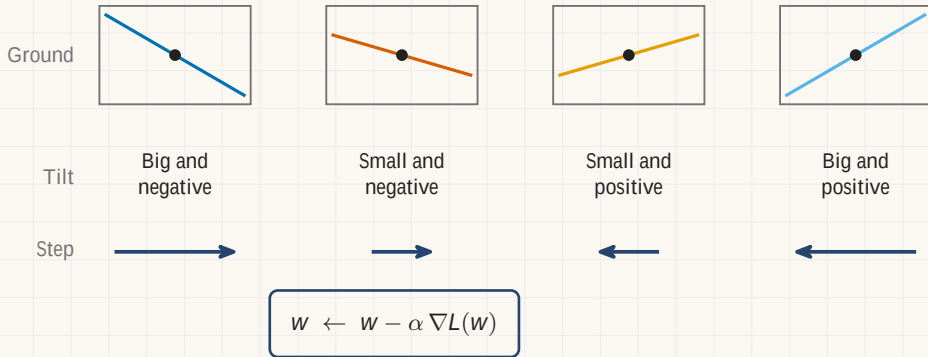
From tilt to step

The same patch of ground, tilted four ways.



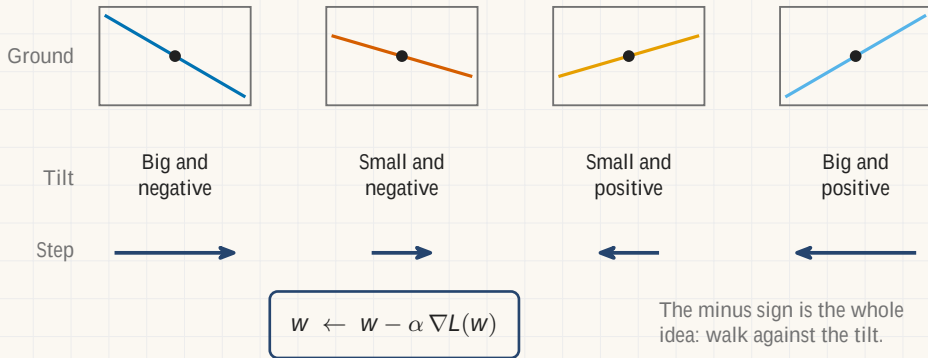
From tilt to step

The same patch of ground, tilted four ways.



From tilt to step

The same patch of ground, tilted four ways.



The algorithm

The 09:00 loop, with the gradient in place of the miss.

```
def gradient_descent(L, w, alpha, eps, tol, steps):  
    for n in range(steps):  
        a = L(w - eps/2)      # height at the left of the window  
        b = L(w + eps/2)      # height at the right of the window  
        g = (b - a) / eps      # the tilt  
        if abs(g) < tol:       # flat enough: you are at the bottom  
            break  
        w = w - alpha * g      # the step: walk against the tilt  
    return w
```

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Our curve, from a bad guess:

```
def L(w):  
    return w**2 - 4*w + 7  
gradient_descent(L, -2, 0.75, 0.1, 0.01, 20)
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gradient_descent(L, -2, 0.75, 0.1, 0.01, 20)
```

eps is the window and tol is how flat counts as flat. alpha is still unchosen.

Weekend 1, Friday morning

A run of gradient descent

$L(w) = w^2 - 4w + 7$, starting at $w = -2$, with $\alpha = 0.75$.

w	$L(w)$	$\nabla L(w)$	$-\alpha \nabla L(w)$	Step
-----	--------	---------------	-----------------------	------

A run of gradient descent

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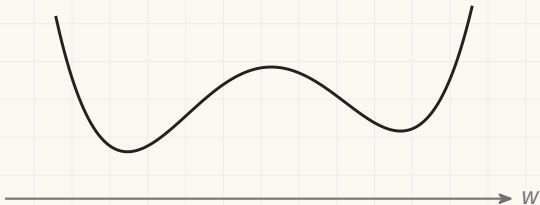
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-2	19	-8	+6	
4	7	4	-3	
1	4	-2	+1.5	
2.5	3.25	1	-0.75	
1.75	3.0625	-0.5	+0.375	

$w^* = 2$, where $L(w^*) = 3$.

The first four rows are the four the room read off by eye. The arrows were the gradient.

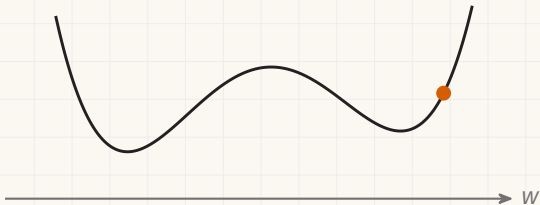
Where gradient descent stops

The rule is local. It only ever knows the ground under its own feet.



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- Every step is downhill.

Where gradient descent stops

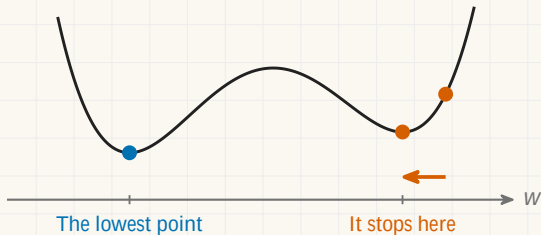
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Where gradient descent stops

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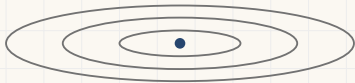
- Every step is downhill.
- The walk stops where the ground is flat.
- Flat is not the same as lowest.

In practice this is answered by starting the walk several times from several places and keeping the best result.

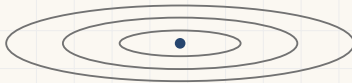
Momentum, 1964*

* Optional. Nothing later in the lecture depends on this frame or the next.

A long narrow valley is where the plain rule is worst. The steep direction sets the step, so the walk crosses the valley instead of following it.



The plain rule

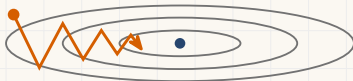


With momentum

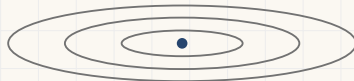
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The plain rule
Eight steps, still short of the bottom

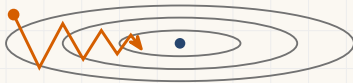


With momentum

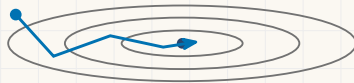
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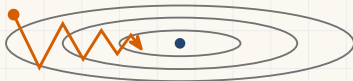


With momentum
Four steps, and it is there

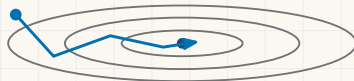
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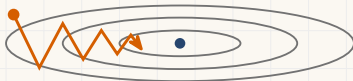
$$v \leftarrow \beta v + \nabla L(w)$$

$$w \leftarrow w - \alpha v$$

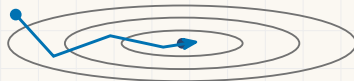
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The plain rule
Eight steps, still short of the bottom



With momentum
Four steps, and it is there

$$\begin{aligned}v &\leftarrow \beta v + \nabla L(w) \\ w &\leftarrow w - \alpha v\end{aligned}$$

v remembers the direction you were already going, at strength β , usually about 0.9. Across the valley it keeps changing sign and cancels; along the floor it keeps its sign and adds up.

Adam, 2015*

* Optional, as the previous frame.

The other failure of one step size: two knobs whose tilts are of completely different size.

One α for both

Knob 1

Knob 2

Each knob on its own scale

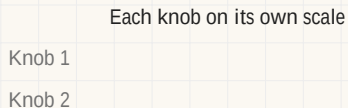
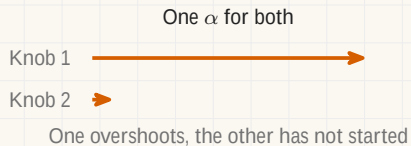
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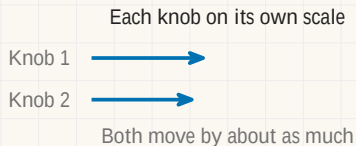
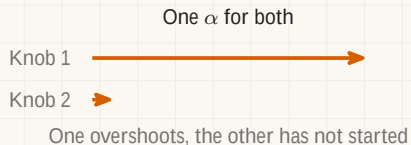
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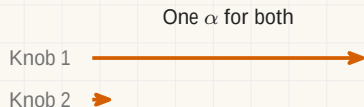
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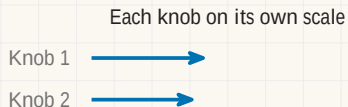
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One overshoots, the other has not started



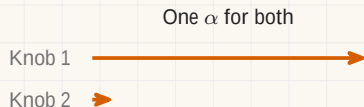
Both move by about as much

$$\begin{aligned}m &\leftarrow \beta_1 m + (1 - \beta_1) \nabla L(w) \\v &\leftarrow \beta_2 v + (1 - \beta_2) \nabla L(w)^2 \\w &\leftarrow w - \alpha m / (\sqrt{v} + \delta)\end{aligned}$$

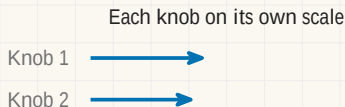
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m is the direction and is exactly momentum. v is the size that knob has been showing. δ keeps the division safe; the paper calls it ϵ and this lecture has spent that letter on the window. The published rule also bias corrects m and v , which matters only for the first few steps.

Weekend 1, Friday morning

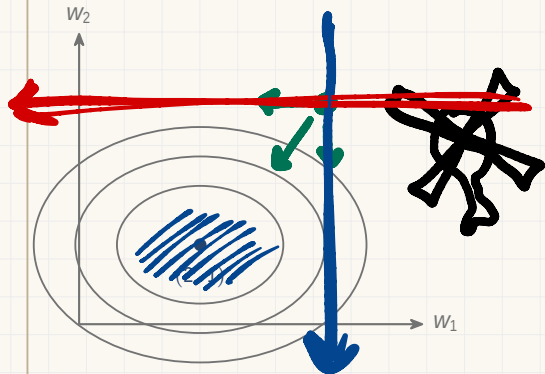
5

Section 5

More than one knob

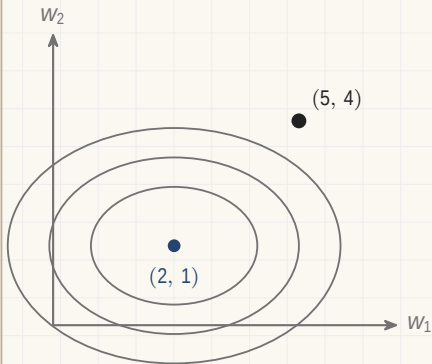
Two knobs

$L(w_1, w_2) = (w_1 - 2)^2 + 2(w_2 - 1)^2 + 3$, seen from above.



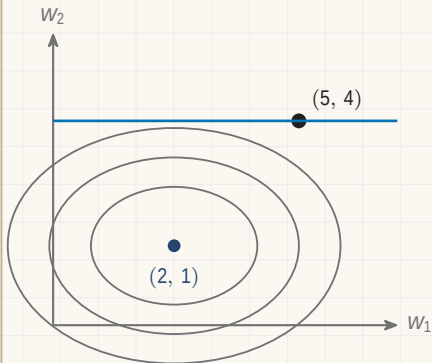
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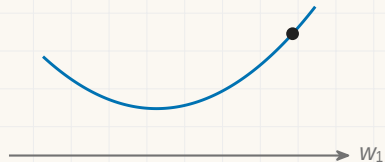


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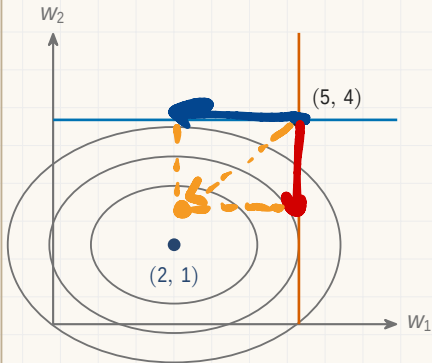


Hold $w_2 = 4$

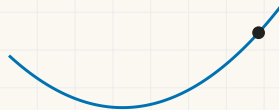


Two knobs

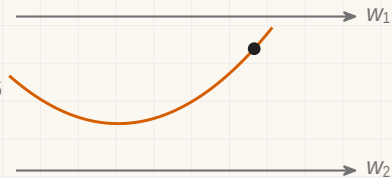
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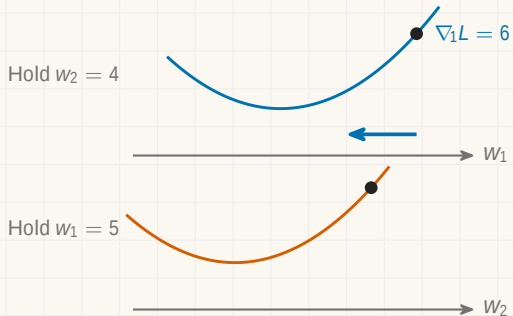
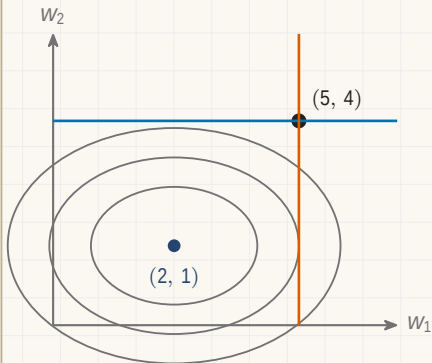


Hold $w_1 = 5$



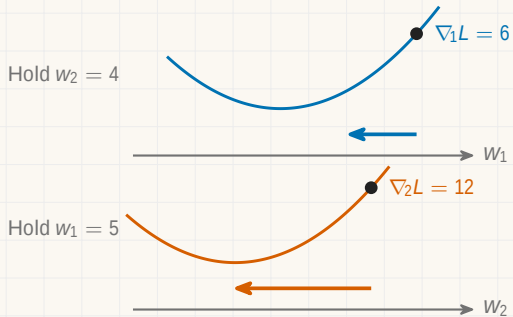
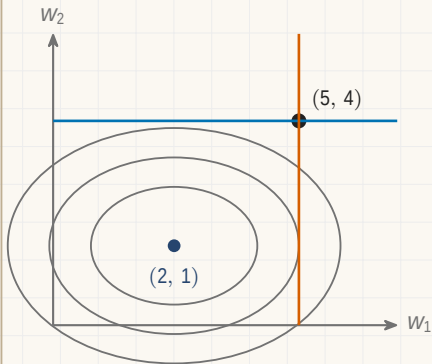
Two knobs

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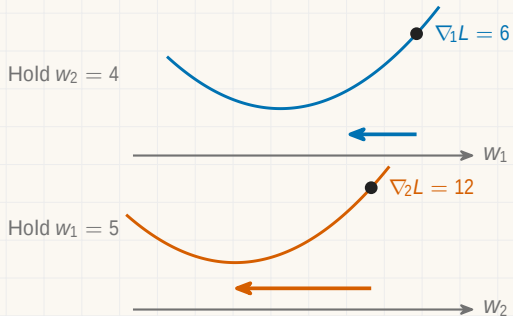
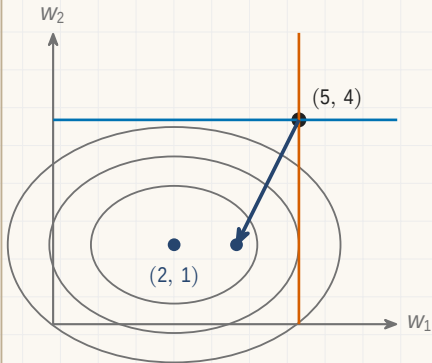
Two knobs

$L(w_1, w_2) = (w_1 - 2)^2 + 2(w_2 - 1)^2 + 3$, seen from above.



Two knobs

$L(w_1, w_2) = (w_1 - 2)^2 + 2(w_2 - 1)^2 + 3$, seen from above.



$$\begin{aligned}(w_1, w_2) &\leftarrow (5, 4) - 0.25 \cdot (6, 12) \\ &= (3.5, 1)\end{aligned}$$

The gradient in d dimensions

One window per knob. Open them one at a time and hold the others still.

Knob

Window

Left

Right

Tilt

Written

The gradient in d dimensions

One window per knob. Open them one at a time and hold the others still.

Knob	Window	Left	Right	Tilt	Written
w_1					
w_2					
$\vdots$					
w_d					

The gradient in d dimensions

One window per knob. Open them one at a time and hold the others still.

Knob	Window	Left	Right	Tilt	Written
w_1	$w_1 \pm \frac{\epsilon}{2}$				
w_2	$w_2 \pm \frac{\epsilon}{2}$				
$\vdots$					
w_d	$w_d \pm \frac{\epsilon}{2}$				

The gradient in d dimensions

One window per knob. Open them one at a time and hold the others still.

Knob	Window	Left	Right	Tilt	Written
w_1	$w_1 \pm \frac{\varepsilon}{2}$	a_1	b_1		
w_2	$w_2 \pm \frac{\varepsilon}{2}$	a_2	b_2		
$\vdots$					
w_d	$w_d \pm \frac{\varepsilon}{2}$	a_d	b_d		

a_j and b_j are the heights at the two edges of knob j 's own window, with every other knob held where it is.

The gradient in d dimensions

One window per knob. Open them one at a time and hold the others still.

Knob	Window	Left	Right	Tilt	Written
w_1	$w_1 \pm \frac{\varepsilon}{2}$	a_1	b_1	$\frac{b_1 - a_1}{\varepsilon}$	
w_2	$w_2 \pm \frac{\varepsilon}{2}$	a_2	b_2	$\frac{b_2 - a_2}{\varepsilon}$	
$\vdots$				$\vdots$	
w_d	$w_d \pm \frac{\varepsilon}{2}$	a_d	b_d	$\frac{b_d - a_d}{\varepsilon}$	

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The gradient in d dimensions

One window per knob. Open them one at a time and hold the others still.

Knob	Window	Left	Right	Tilt	Written
w_1	$w_1 \pm \frac{\varepsilon}{2}$	a_1	b_1	$\frac{b_1 - a_1}{\varepsilon}$	$\nabla_1 L(w)$
w_2	$w_2 \pm \frac{\varepsilon}{2}$	a_2	b_2	$\frac{b_2 - a_2}{\varepsilon}$	$\nabla_2 L(w)$
$\vdots$				$\vdots$	$\vdots$
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a_j and b_j are the heights at the two edges of knob j 's own window, with every other knob held where it is.

The gradient in d dimensions

One window per knob. Open them one at a time and hold the others still.

Knob	Window	Left	Right	Tilt	Written
w_1	$w_1 \pm \frac{\varepsilon}{2}$	a_1	b_1	$\frac{b_1 - a_1}{\varepsilon}$	$\begin{bmatrix} \nabla_1 L(w) \\ \nabla_2 L(w) \\ \vdots \\ \nabla_d L(w) \end{bmatrix}$
w_2	$w_2 \pm \frac{\varepsilon}{2}$	a_2	b_2	$\frac{b_2 - a_2}{\varepsilon}$	
$\vdots$				$\vdots$	
w_d	$w_d \pm \frac{\varepsilon}{2}$	a_d	b_d	$\frac{b_d - a_d}{\varepsilon}$	

a_j and b_j are the heights at the two edges of knob j 's own window, with every other knob held where it is.

$$\nabla L(w) = (\nabla_1 L(w), \dots, \nabla_d L(w))$$

Gradient descent in d dimensions

The same rule. The guess is a list, the gradient is a list, and they subtract term by term.

$$w \leftarrow w - \alpha \nabla L(w), \quad w \in \mathbb{R}^d$$

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Coordinate by coordinate, that is d separate copies of the one dimensional rule:

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```
for j in range(d):  
    a = L(w_shift(w, j, -eps/2))    # move knob j back by eps/2  
    b = L(w_shift(w, j, +eps/2))    # move knob j on by eps/2  
    grad[j] = (b - a) / eps  
w = [w[j] - alpha * grad[j] for j in range(d)]
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That costs $2d$ evaluations of L per step, and for a neural network d runs to millions.

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```

That costs $2d$ evaluations of L per step, and for a neural network d runs to millions.

The right way is backpropagation, which is the 14:00 lecture.